ProprioVLA: Self-Supervised Proprioceptive Grounding for Vision-Language-Action Policies

Anonymous Author(s)
Affiliation
Address
email

Abstract

Vision-language-action (VLA) policies have shown strong promise as generalist controllers for robot manipulation, yet their action predictions can remain fragile under changes in visual perspective. We identify a systematic failure pattern that lateral camera shifts induce co-directional shifts in predicted robot trajectories, suggesting that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. To bridge this gap, we introduce PROPRIOVLA, a self-supervised alignment approach that incentivizes self-grounding capability in VLA policies. PROPRIOVLA implicitly encourages the policy to localize its own embodiment by aligning learned grounding queries with state-aligned residual flow targets mined from robot interaction. Specifically, PROPRIOVLA turns proprioceptively aligned visual change, captured by optical flow and image difference, into token-level self-grounding targets. It then trains a lightweight learned-query head over visual tokens in the VLA representation to recover these targets with normalized KL supervision, regularizing the visual representation to encode the visible embodiment without manual masks, external grounding datasets, or stronger foundation-model supervision. Extensive experiments on both real-robot and simulation environments show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, suggesting that proprioceptive grounding is a practical route toward spatially reliable VLA control.

1 Introduction

Vision-language-action (VLA) policies are becoming a promising route toward general-purpose robot manipulation. By scaling visual-language pretraining and robot demonstrations, recent robot foundation models can follow language instructions and transfer across tasks and embodiments [2, 3, 7, 5]. Yet spatially reliable control requires a capability that is easy to overlook: the policy must ground the action not only to the instructed object, but also to the body that executes the action. We refer to this capability as *self-grounding*, the binding between proprioceptive state and visual evidence of the robot's own gripper or end-effector.

We make this missing requirement concrete with two diagnostic findings. First, when the physical robot-scene state is fixed and only the camera viewpoint is laterally shifted, VLA-predicted trajectories shift in the same image direction. Second, even robot-centric prompts such as "Where is the robot itself?" often produce attention maps that fail to localize the gripper or end-effector. Together, these findings reveal that task grounding does not imply self-grounding: a policy may identify the instructed object while failing to anchor its action prediction to the visible body that will execute it.

This failure suggests that the action frame learned by current VLA adaptation can remain entangled with the image frame. Standard imitation or diffusion action losses reward correct trajectories under

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

the training observation distribution, but they do not require the visual representation to encode where the robot appears. Proprioception may be available as an input, yet it can act as a separate state cue while visual tokens exploit pixel-coordinate shortcuts, camera-specific correlations, or object-location priors. Such policies can therefore appear competent in-distribution while exhibiting co-directional trajectory drift under camera-only perturbations.

Recent spatial and grounding-oriented VLA methods make important progress, but they do not directly provide a self-supervised signal for self-grounding. Existing approaches improve spatial reasoning through visual trajectory prompts, 3D representations, camera-space action formulations, view synthesis, or explicit treatments of proprioception [11, 8, 9, 4, 10, 6]. These methods make task space, viewpoint, or state more explicit, yet they primarily supervise task execution rather than the visual binding between proprioceptive change and the acting body. What is missing is a training signal that tells the VLA where its own visible embodiment is, without manual robot-part masks, external grounding datasets, or stronger teacher models.

PROPRIOVLA addresses this gap by turning robot interaction itself into self-grounding supervision. The key observation is that visual change from the acting body is synchronized with proprioceptive change. Temporal image differences and optical flow capture residual visual change, while proprioceptive changes indicate when that change should correspond to the robot body rather than unrelated background motion. By aligning these two sources, PROPRIOVLA constructs confidence-weighted token-level self-grounding targets over the visual-token grid using only observations and states already present in robot demonstrations.

Building on these targets, PROPRIOVLA trains a lightweight learned-query grounding head over the VLA visual representation. The head reads visual tokens through cross-attention and predicts a self-grounding distribution, which is supervised to recover the state-aligned targets with a normalized KL objective. Importantly, this branch is auxiliary: the mined targets are used only during training, the action path never consumes them, and the policy interface is unchanged at inference. The effect is to regularize shared visual tokens toward visible embodiment while preserving the standard action-supervised VLA objective.

We evaluate PROPRIOVLA in simulation and real-robot manipulation, including controlled camera-perturbation probes that isolate view-induced action drift. Across settings, PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation. Ablations further show that state-aligned target mining outperforms raw motion saliency alone, and that normalized KL provides a stable distribution-matching objective across visual token counts and target sharpness.

Our contributions are:

- We identify a missing-self failure mode in VLA policies: camera-only viewpoint shifts can induce co-directional shifts in predicted robot trajectories, while robot-centric prompting can still fail to localize the gripper or end-effector.
- We introduce State-Aligned Residual Flow (SARF), which turns proprioceptively aligned optical-flow and image-difference cues into confidence-weighted token-level self-grounding targets without manual masks, external grounding datasets, or stronger foundation-model supervision.
- We propose PROPRIOVLA, a self-supervised proprioceptive alignment approach that trains a learned-query grounding head with normalized KL supervision, improving task success and robustness in real-robot and simulation environments.

2 Related Work

Vision-language-action models. Vision-language-action (VLA) policies scale language-conditioned robot control by coupling vision-language backbones with robot action prediction. RT-1 and RT-2 established the value of large robot datasets and web-scale semantic priors [2, 3]; recent open and generalist models further improve transfer across tasks, embodiments, and action parameterizations [7, 5, 1]. However, this line of work mainly optimizes task success under the training observation distribution. It does not directly test whether the action-relevant visual evidence is anchored to camera-invariant robot-object geometry. PROPRIOVLA studies this missing failure mode: under camera perturbations, a fine-tuned VLA can follow image-coordinate shortcuts instead of grounding actions in the robot's own state.

Visual grounding in VLA policies. Recent work improves VLA perception by making space, viewpoint, or proprioception more explicit. TraceVLA adds visual trajectory prompts to increase spatial-temporal awareness [11]; SpatialVLA injects 3D spatial representations [8]; OC-VLA predicts actions in camera space to reduce observation-action mismatch [9]; and VistaBot uses geometry-aware view synthesis for cross-view robustness [4]. These methods show that perception is a central bottleneck, but they depend on explicit prompts, 3D encodings, camera coordinate transforms, or generative view modules. A complementary line questions proprioception itself: State-free Policy removes proprioceptive inputs to avoid trajectory overfitting [10], while GAP shows that proprioceptive gradients can suppress visual learning during motion transitions [6]. PROPRIOVLA takes a different stance: it does not discard proprioception or treat it as a passive token. Instead, it uses proprioceptive change to mine token-level self-grounding targets, turning the robot's own motion into self-supervision for learned visual attention.

3 Does a VLA Know Where It Is?

A manipulation policy must know where it is, not only what object the language instruction names. We probe this self-location requirement at two levels. The action-level probe asks whether a predicted trajectory remains tied to the fixed robot-scene state when only the camera view is shifted. The representation-level probe asks whether robot-relevant attention localizes the arm, gripper, or end-effector rather than only task objects or background context. Together, these probes expose a *missing-self* failure mode: a policy can appear task-grounded while lacking a stable visual grounding of the body that acts.

3.1 Finding 1: The Action Frame Is Entangled with the Image Frame

We diagnose action-frame grounding by perturbing the view while holding the physical manipulation state fixed. Given observation o_t, instruction x, proprioceptive state s_t, and policy π_θ, let $\tau_\theta = \pi_\theta(o_t, x, s_t)$ be the predicted trajectory from the original view. We then apply a lateral camera or image shift $T_{\Delta v_{\mathrm{cam}}}$ and compute $\tau'_\theta = \pi_\theta(T_{\Delta v_{\mathrm{cam}}}(o_t), x, s_t)$. The instruction, robot state, object configuration, and physical scene are unchanged; only the image frame moves.

If the action frame were grounded in robot-scene geometry, the prediction would remain stable under this camera-only perturbation, up to the known transform used for visualization. A missing-self failure appears when the predicted end-effector displacement follows the image displacement. We quantify this effect after projecting the trajectory change onto a common lateral coordinate system:

$$\mathrm{TSR} = \frac{\|\Delta\tau_{\mathrm{pred}}\|_2}{\|\Delta v_{\mathrm{cam}}\|_2 + \epsilon},$$
$$\mathrm{SDA} = \frac{\langle \Delta\tau_{\mathrm{pred}}, \Delta v_{\mathrm{cam}} \rangle}{\|\Delta\tau_{\mathrm{pred}}\|_2 \|\Delta v_{\mathrm{cam}}\|_2 + \epsilon}.$$

High TSR means the prediction changes strongly under a view-only shift; high SDA means the change is co-directional with the image shift. The combination indicates that the learned action frame is partially entangled with the image frame rather than fully anchored to proprioceptive state and robot-object geometry.

3.2 Finding 2: Task Grounding Does Not Imply Self-Grounding

We next test whether action-relevant representations localize the acting body. For a VLA or its vision-language backbone, we use robot-centric prompts such as "Where is the robot arm?", "Please identify the robot itself in the scene," and "Where is the manipulator?" We then visualize attention heatmaps, attention rollout, Grad-CAM maps, cross-attention maps, or token relevance maps. The same analysis can be applied to policy tokens used for action prediction, comparing task relevance with self relevance.

This probe separates task grounding from self-grounding. A model may attend to task objects, the tabletop, or salient background regions while assigning little relevance to the robot arm, gripper, or end-effector. Such attention can support object recognition and instruction following, but it does not bind the action to the body that will execute it. Robust VLA control therefore requires binding

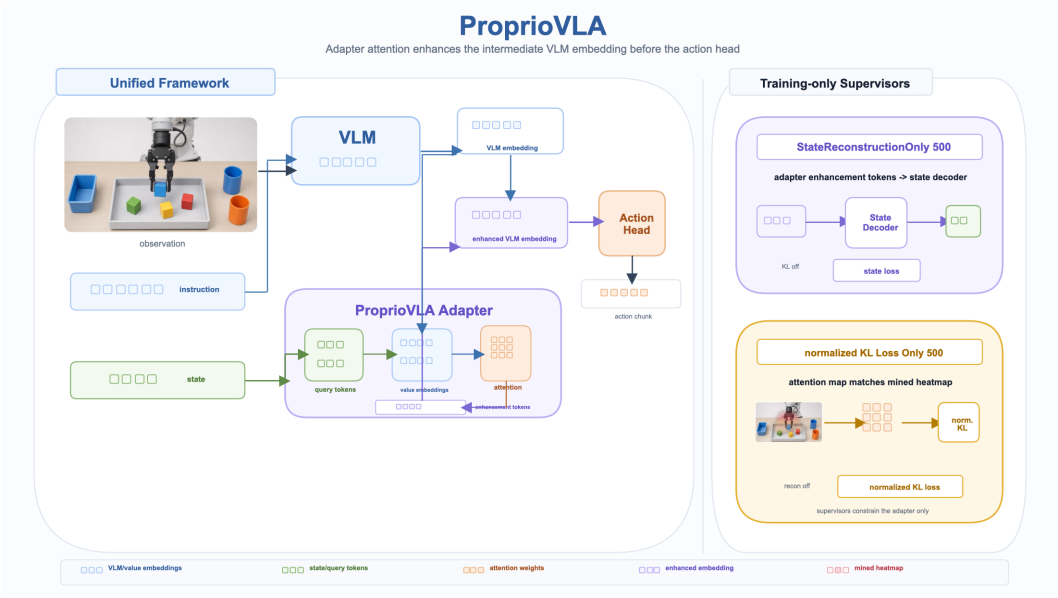

Figure 1: Overview of PROPRIOVLA. Given an observation, instruction, and proprioceptive state, the base VLA produces a standard multimodal representation. State-Aligned Residual Flow (SARF) mines token-level self-grounding targets from proprioceptively aligned visual change, while the Proprioceptive Alignment Module (PAM) trains learned grounding queries over VLA visual tokens to recover these targets. The mined target distribution is used only during training.

proprioceptive state to visual evidence of the acting body, rather than treating proprioception as a passive input that the visual pathway can ignore.

4 PROPRIOVLA

PROPRIOVLA adds a self-supervised proprioceptive grounding objective to a VLA policy without changing the policy's action interface. Given an observation o_t, language instruction x, proprioceptive state s_t, and action label a_t, the base VLA produces multimodal tokens and is trained with its standard imitation or diffusion action loss. PROPRIOVLA adds two training-time components: State-Aligned Residual Flow (SARF), which mines token-level self-grounding targets from robot interaction, and a Proprioceptive Alignment Module (PAM), which predicts a self-grounding distribution over VLA visual tokens. The pseudo-targets and grounding loss are used only during training; at inference, the policy consumes the same inputs as the base VLA.

4.1 Problem Setup and Visual Tokens

We formulate proprioceptive grounding as an auxiliary distribution-matching problem over the VLA visual-token grid. Let the VLA backbone return a sequence of hidden states $Z_t = \{z_{t,i}\}_{i=1}^{L}$. An image-token mask selects the subset of tokens that originate from visual inputs, and the image-token layout records the spatial grid shape for each view. For the target camera view, we collect the valid visual tokens

$$V_t = \{v_t(p) \in \mathbb{R}^d\}_{p \in \mathcal{P}_t},$$

where $\mathcal{P}_t$ indexes valid spatial token locations. Each token has a normalized center coordinate $u_p = (x_p, y_p) \in [0,1]^2$, computed from the corresponding image grid. If multiple image units are present, we group units by view index and apply the grounding objective to the selected view; invalid or missing views are ignored by the token mask.

The objective is to make the VLA representation recover a distribution over where the acting body appears. During training, SARF provides a target distribution H_t over $\mathcal{P}_t$. PAM predicts a distribution A_t from the VLA visual tokens. The grounding loss supervises A_t to match H_t, while gradients flow

through the same visual representation used by action prediction. This design separates grounding supervision from action execution: grounding is learned as a side objective, not as an additional runtime input.

4.2 State-Aligned Residual Flow

SARF mines self-grounding targets from motion that is visually salient and proprioceptively supported. For an episode, we first compute a scalar proprioceptive activity signal from end-effector translation and gripper change,

$$\gamma_t = \mathrm{norm}\big(\|\Delta r_t\|_2 + \beta|\Delta g_t|\big),$$

where r_t is the end-effector position, g_t is the gripper state, β balances gripper activity, and $\mathrm{norm}(\cdot)$ denotes robust percentile normalization. High γ_t indicates frames where the robot state changes enough to make visual self-motion informative.

SARF next estimates residual visual change from consecutive frames. Let I_t be the grayscale image at time t, and let $F_{t\to t+1}$ be dense optical flow between adjacent frames. We combine temporal image difference and optical-flow magnitude into a visual-change score

$$M_t = \mathcal{N}\left(\frac{1}{2}\left(|I_t - I_{t-1}| + |I_{t+1} - I_t|\right) + \frac{1}{4}\left(\|F_{t-1\to t}\|_2 + \|F_{t\to t+1}\|_2\right)\right),$$

where $\mathcal{N}$ denotes map normalization followed by mild Gaussian smoothing. Image difference captures appearance changes, while optical flow captures coherent displacement. Their combination suppresses static background tokens and highlights image regions whose change can plausibly correspond to the robot or objects directly coupled to it.

To reduce spurious motion from distractors, SARF fits a state-guided spatial prior from high-confidence motion peaks within each episode. A lightweight weighted regression maps proprioceptive state features to image points, using motion-derived confidence as weights and robust inlier support to reject outliers. When the fit is reliable, it predicts a prior point μ_t and support score q_t for each frame. SARF combines this prior with visual change as

$$S_t(p) = \mathcal{N}\left(M_t(p)^{0.85}\left[(1-\rho_t) + \rho_t G(p;\mu_t,\sigma_{\mathrm{prior}})\right]\right),$$

where G is a Gaussian point prior and ρ_t is capped by the fitted model quality. If the state-guided fit is unreliable, SARF falls back to the pure motion score M_t. This fallback keeps target mining self-supervised and avoids forcing a poor state-to-image fit onto ambiguous episodes.

SARF converts the score map into a token-level target. We blur S_t, extract a peak region from the high-score set, and obtain a normalized pseudo point y_t, anisotropic scale $\sigma_t = (\sigma_t^x, \sigma_t^y)$, bounding box B_t, and confidence c_t. The confidence is a geometric combination of peak sharpness, visual-change energy, proprioceptive activity, and, when available, state-prior support. Low-confidence frames are abstained from by setting $c_t = 0$ below a minimum threshold.

Given token centers $\{u_p\}_{p\in\mathcal{P}_t}$, we rasterize the mined pseudo point into a normalized target distribution:

$$\tilde{H}_t(p) = \exp\left(-\frac{1}{2}\left[\left(\frac{u_p^x - y_t^x}{\sigma_t^x}\right)^2 + \left(\frac{u_p^y - y_t^y}{\sigma_t^y}\right)^2\right]\right)\mathbb{1}[u_p \in B_t],$$

$$H_t(p) = \frac{\tilde{H}_t(p)}{\sum_{q\in\mathcal{P}_t}\tilde{H}_t(q) + \epsilon}.$$

If the clipped box contains no valid token, we fall back to the unclipped Gaussian target, and if that also has no mass, to the uniform distribution over valid tokens. Thus SARF produces a stable token-level self-grounding target from interaction alone, without robot-part masks or external grounding labels.

4.3 Proprioceptive Alignment Module

PAM predicts the self-grounding distribution with a lightweight learned-query head over VLA visual tokens. The module maintains M trainable grounding latents $Q = \{q_m\}_{m=1}^{M}$. In the main learned-query setting, these latents are independent parameters; a state-conditioned query variant, used for

ablation, adds a projection of the grounding state to the same latents. Each visual token is augmented with a learned positional embedding of its normalized center coordinate u_p.

PAM applies a small stack of cross-attention blocks from grounding latents to visual tokens. For one block, with token inputs $\hat{v}_t(p)$, the attention logits are

$$\ell_t(m,p) = \bar{q}_m^\top \bar{k}_t(p), \qquad \bar{q}_m = \mathrm{norm}(W_q\,\mathrm{LN}(q_m)), \quad \bar{k}_t(p) = \mathrm{norm}(W_k\,\mathrm{LN}(\hat{v}_t(p))).$$

The block uses the corresponding softmax attention to update the grounding latents through value projection and a feed-forward layer. The final-layer attention logits define the predicted self-grounding distribution:

$$A_t(p) = \frac{1}{M}\sum_{m=1}^{M} \mathrm{softmax}_{q\in\mathcal{P}_t}\big(\ell_t(m,q)\big)_p.$$

Averaging across latents lets several queries cover the same visible embodiment while still producing one distribution for supervision.

PAM is intentionally a side head rather than a token-writeback module. The grounding branch reads VLA visual tokens and returns attention logits for training, but the original VLA token sequence and attention mask are passed to the action decoder unchanged. In the normalized-KL-only setting, state reconstruction and feature reconstruction auxiliaries are disabled, and the grounding loss acts through the attention-producing query/key pathway. This keeps the method faithful to the central goal: shaping shared visual tokens to encode visible embodiment, without adding pseudo-label-dependent inputs to the policy at test time.

4.4 Normalized KL Grounding Objective

The grounding objective compares the predicted distribution A_t to the SARF target H_t with a scale-normalized KL divergence. Let

$$\mathrm{CE}_t = -\sum_{p\in\mathcal{P}_t} H_t(p)\log A_t(p), \qquad \mathcal{H}(H_t) = -\sum_{p\in\mathcal{P}_t} H_t(p)\log H_t(p).$$

The KL from target to model is $\mathrm{CE}_t - \mathcal{H}(H_t)$. Because its magnitude depends on token count and target sharpness, we normalize it by the KL from the target to uniform attention:

$$\kappa_t = \frac{\mathrm{CE}_t - \mathcal{H}(H_t)}{\log|\mathcal{P}_t| - \mathcal{H}(H_t) + \epsilon}.$$

With this normalization, a target-matching prediction approaches zero, while uniform attention is approximately one. This makes the auxiliary loss stable across different image-token counts, pseudo-box sizes, and target sharpness.

The complete normalized-KL-only training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{act}} + \lambda_g \frac{\sum_t \mathbb{1}[c_t \geq c_{\min}] c_t \kappa_t}{\sum_t \mathbb{1}[c_t \geq c_{\min}] c_t + \epsilon}.$$

$\mathcal{L}_{\mathrm{act}}$ is the original VLA action loss. The implementation keeps $\mathcal{L}_{\mathrm{act}}$ active in all grounding modes; in normalized-KL-only mode, it forces the localization auxiliary to use normalized KL and sets the state-reconstruction and feature-reconstruction weights to zero. Since normalized KL is already calibrated against the uniform baseline, it is not further rescaled by action-loss balancing. The result is a training signal that teaches visual self-grounding while preserving the standard VLA inference path.

5 Experiments

The experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. The evaluation links three evidence types: task performance, action stability under camera perturbations, and visual grounding of the acting robot.

5.1 Experimental Setup

We evaluate a VLA backbone fine-tuned on language-conditioned manipulation demonstrations in simulation and on a real robot. Each episode contains visual observations o_t, proprioceptive state s_t, language instruction x, and action supervision a_t. For robustness evaluation, we keep the physical robot-scene state, instruction, and proprioceptive state fixed, then apply lateral view shifts of controlled magnitude. This isolates camera-induced changes from changes in the underlying manipulation state.

We compare four policies under the same demonstrations, action labels, and view-shift protocol: Base VLA, action-supervised fine-tuning, VLA with proprioception concatenation, and PROPRIOVLA. We report task success rate, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and a grounding score. The grounding score is computed as either attention-target alignment with H_t or attention mass on robot regions. Lower trajectory shift and direction agreement indicate a more stable action frame; higher grounding score indicates stronger localization of the acting body. Each setup reports the task suite, demonstration count, train/test split, camera shifts, action parameterization, proprioceptive normalization, optimizer, and the grounding loss weight λ_g.

5.2 Simulation Experiments

Simulation provides controlled access to camera shifts, object layouts, and ground-truth robot geometry. We evaluate normal, mild-shift, medium-shift, and large-shift settings. For each setting, we report task success, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and grounding score. Robustness curves plot these metrics against shift magnitude, making the co-movement between predicted actions and image shifts directly visible.

We also evaluate held-out object layouts and camera positions. These splits separate memorization of a fixed training camera from grounding that transfers across views. Qualitative trajectory overlays compare the original prediction with shifted-view predictions after mapping them to the same robot-centric coordinate frame.

5.3 Real-World Experiments

The real-robot evaluation tests whether the same failure mode appears under sensor noise, calibration error, and real scene variation. We evaluate several manipulation tasks under the original view and shifted views, reporting task success, success drop under perturbation, action prediction error on logged demonstrations, and grounding score.

For qualitative analysis, we visualize baseline attention, the mined state-aligned target distribution, PROPRIOVLA attention, and trajectory overlays for the same scene. The comparison checks whether attention remains dominated by task objects and background context or shifts toward the arm, gripper, and end-effector while the trajectory remains stable.

5.4 Ablations

The ablation study isolates the mechanisms that distinguish PROPRIOVLA from simple state conditioning. We evaluate four variants: (1) removing the visual self-grounding loss $\mathcal{L}_{\text{ground}}$; (2) replacing normalized KL with an unnormalized localization loss; (3) using raw optical flow instead of state-aligned residual motion; and (4) replacing the learned grounding queries with state-derived queries. Each ablation is evaluated with the same success, action, trajectory-shift, and grounding metrics.

These ablations test the central mechanism: the policy must use proprioception to mine visual self-grounding targets, rather than merely receive proprioceptive values as extra inputs. The comparison separates three alternatives: gains from additional state, gains from motion saliency alone, and gains from learned-query self-grounding.

6 Limitations

PROPRIOVLA assumes that the robot is at least partially visible during training. If the embodiment is consistently outside the camera view, the state-aligned motion signal cannot provide useful self-

grounding supervision, and the visual query has no direct evidence from which to recover the robot's own state.

State-aligned residual motion can be ambiguous when non-robot motion is coupled to the robot. Grasped objects, deformable materials, moving backgrounds, or other agents may move synchronously with proprioceptive change and receive high alignment scores. Confidence weighting reduces weak supervision, but it cannot fully distinguish the robot body from every object moved by that body.

Finally, the evaluation covers a finite set of embodiments, camera perturbations, and manipulation tasks. Broader multi-camera setups, mobile manipulation, multi-robot scenes, long-horizon tasks, and highly dynamic environments remain important tests for whether proprioceptive self-grounding scales beyond controlled perturbation regimes.

7 Conclusion

This paper studies a missing-self failure mode in VLA policies: under camera-only viewpoint shifts, predicted robot trajectories can move with the image frame even when the physical robot-scene state is unchanged. The diagnostic shows that task grounding and self-grounding are distinct requirements for spatially reliable manipulation.

PROPRIOVLA addresses this issue through two connected components: State-Aligned Residual Flow (SARF) and a Proprioceptive Alignment Module (PAM). SARF mines token-level self-grounding targets from robot interaction without manual masks or external grounding datasets, and PAM trains learned grounding queries to recover these targets with normalized KL supervision. Learning to see the self is a practical step toward VLA policies whose actions are grounded in embodiment rather than in the image frame alone.

References

[1] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0: A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.

[2] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Jasmine Hsu, et al. RT-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.

[3] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, 2023.

[4] Songen Gu, Yuhang Zheng, Weize Li, Yupeng Zheng, Yating Feng, Xiang Li, Yilun Chen, Pengfei Li, and Wenchao Ding. VistaBot: View-robust robot manipulation via spatiotemporal-aware view synthesis. *arXiv preprint arXiv:2604.21914*, 2026.

[5] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[6] Jingxian Lu, Wenke Xia, Yuxuan Wu, Zhiwu Lu, and Di Hu. When would vision-proprioception policies fail in robotic manipulation? *arXiv preprint arXiv:2602.12032*, 2026.

[7] Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, et al. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024.

[8] Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, and Xuelong Li. SpatialVLA: Exploring spatial representations for visual-language-action model. *Robotics: Science and Systems*, 2025.

[9] Tianyi Zhang, Haonan Duan, Haoran Hao, Yu Qiao, Jifeng Dai, and Zhi Hou. Grounding actions in camera space: Observation-centric vision-language-action policy. *arXiv preprint arXiv:2508.13103*, 2025.

[10] Juntu Zhao, Wenbo Lu, Di Zhang, Yufeng Liu, Yushen Liang, Tianluo Zhang, Yifeng Cao, Junyuan Xie, Yingdong Hu, Shengjie Wang, et al. Do you need proprioceptive states in visuomotor policies? *arXiv preprint arXiv:2509.18644*, 2025.

[11] Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. TraceVLA: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. *arXiv preprint arXiv:2412.10345*, 2025.

A Additional Experimental Details

Put supplementary derivations, extra implementation details, additional tables, and qualitative examples here.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.